

Notes

Assignments

◆ 4.14, 4.25, 4.31, 4.33, 4.35

Tutorial problems

- Basic Problems with Answers 4.10, 4.16
- Basic Problems 4.26, 4.30

Mid-term examination

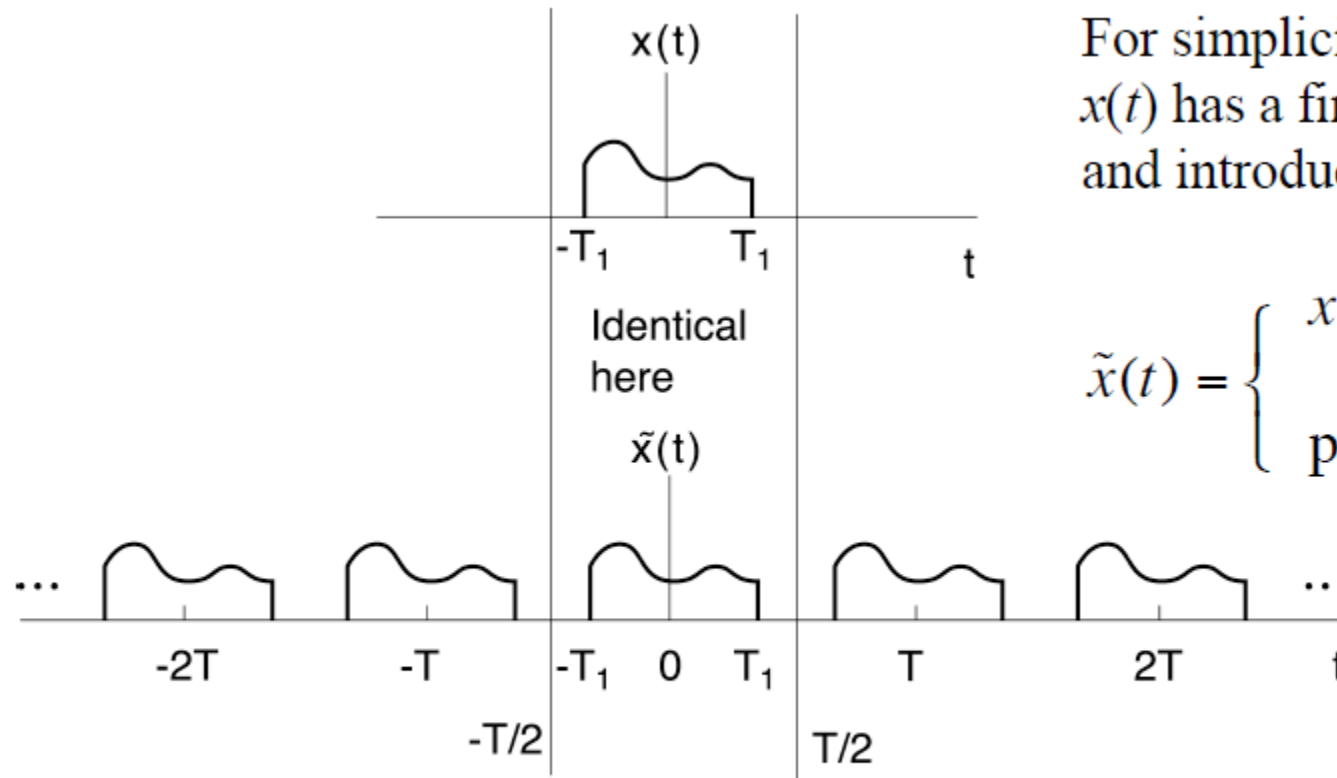
- Time: 11-17-2024 (Sunday) 7:00-9:00 pm
- Venue: 1教
- Range: Chapters 1-4
- Problem language: English
- Final examination: 40% for Chapters 1-4
- Basic equations/formulas are provided

Chapter 4

The Continuous-Time Fourier Transform

(cont.)

So, on the derivation of FT ...



For simplicity, assume $x(t)$ has a finite duration, and introduce a periodic $\tilde{x}(t)$

$$\tilde{x}(t) = \begin{cases} x(t) & -\frac{T}{2} < t < \frac{T}{2} \\ \text{periodic} & |t| > \frac{T}{2} \end{cases}$$

As $T \rightarrow \infty$, $x(t) = \tilde{x}(t)$ for all t

The CT Fourier Transform Pair

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt \quad \text{— } FT$$

Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega \quad \text{— Inverse } FT$$

Inverse Fourier Transform

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

CT Fourier Transform Pair

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega$$

CT Fourier Series Pair

$$a_k = \frac{1}{T} \int_T x(t)e^{-jk\omega_0 t} dt$$

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

Harmonically related

With CTFT, now the frequency response of an LTI system makes complete sense

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} H(jk\omega_0) a_k e^{jk\omega_0 t}$$

$$a_k \longrightarrow \underbrace{H(jk\omega_0) a_k}_{\text{"gain"}}$$

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t) e^{-j\omega t} dt$$



Impulse response $\xleftrightarrow{\mathcal{F}}$ Frequency response

CT Fourier Transforms of **Periodic** Signals

Suppose

$$X(j\omega) = \delta(\omega - \omega_0)$$

$\Downarrow$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t}$$

— periodic in t with frequency ω_0

That is

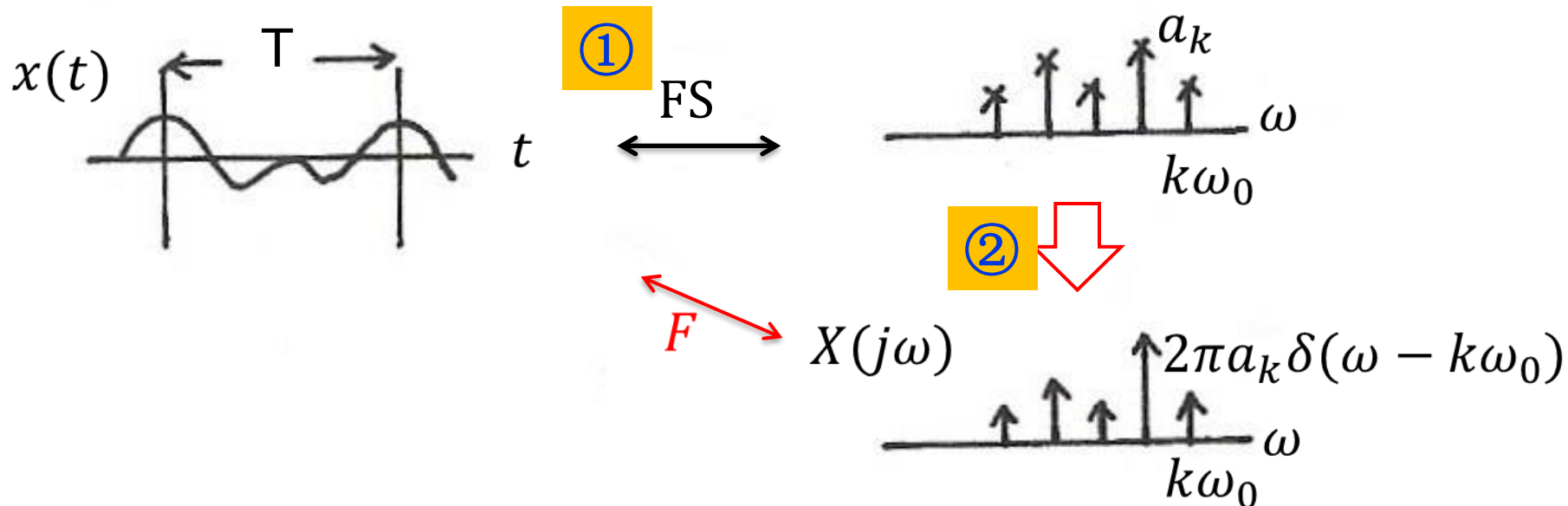
$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

— All the energy is concentrated in one frequency — ω_0

Fourier Transform for Periodic Signals – Unified Framework

More generally, if $x(t) = x(t+T)$, then

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \longleftrightarrow X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0) \quad \text{Discrete spectra}$$

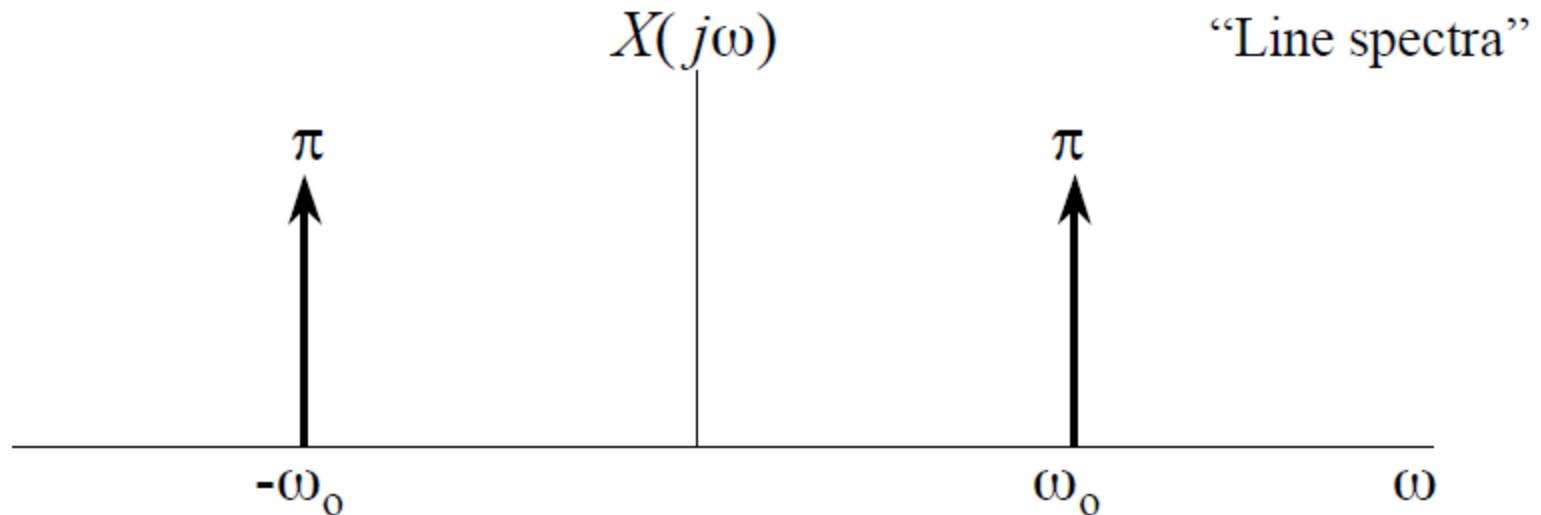


Example 4.7

$$x(t) = \cos \omega_0 t = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}$$



$$X(j\omega) = \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$$



How about $\sin \omega_0 t$?

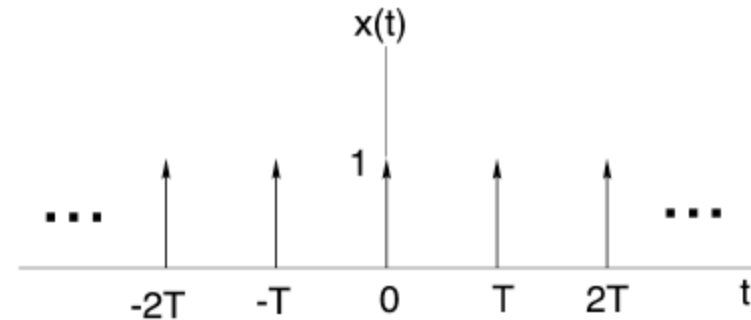
Example 4.8

$$x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad \text{— sampling function}$$

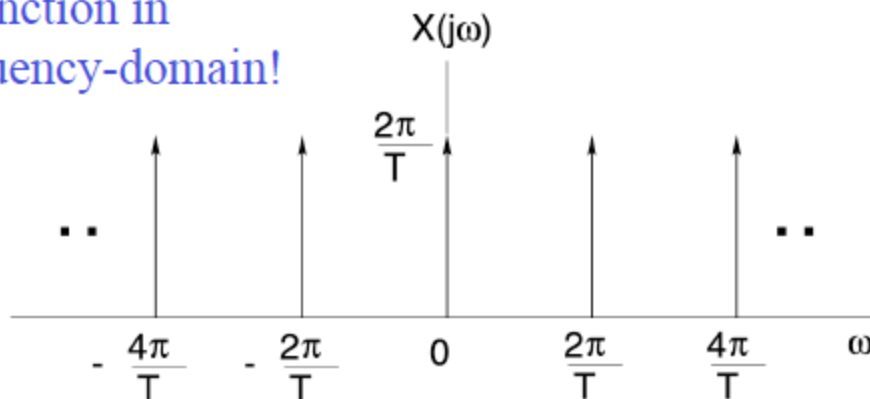
$$x(t) \xleftrightarrow{\text{FS}} a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_o t} dt = \frac{1}{T}$$

$$\Downarrow x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} = \sum_{k=-\infty}^{+\infty} \frac{1}{T} e^{jk\omega_o t}$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2\pi}{T} \delta(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_o})$$



Same function in
the frequency-domain!

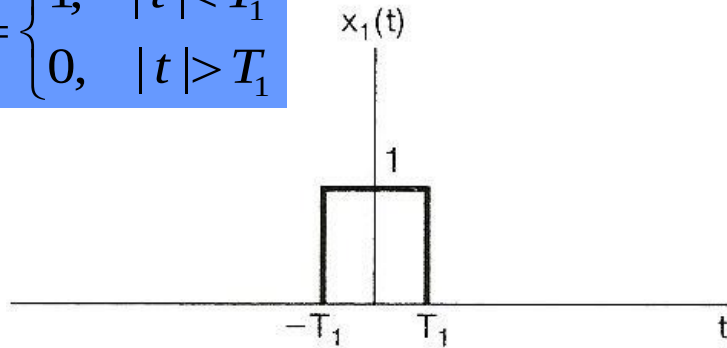


Note in this case, periodic
in both time domain (with
a period T) and frequency
domain (with a period
 $2\pi/T$)

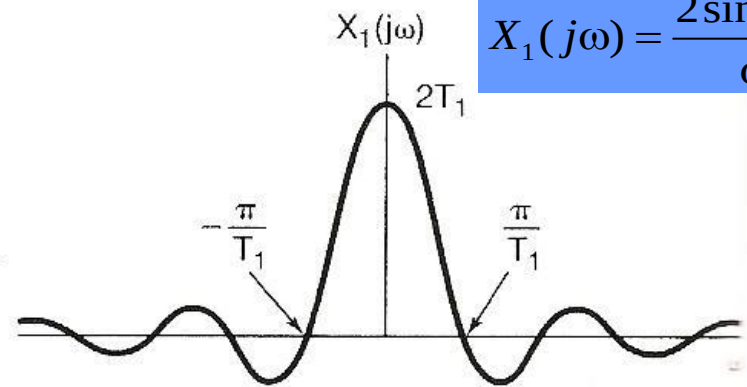
CTFT Properties (cont.)

6) Duality

$$x_1(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$$

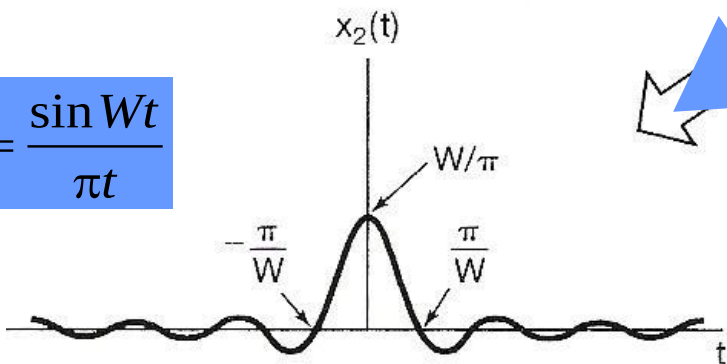


$\mathcal{F}$

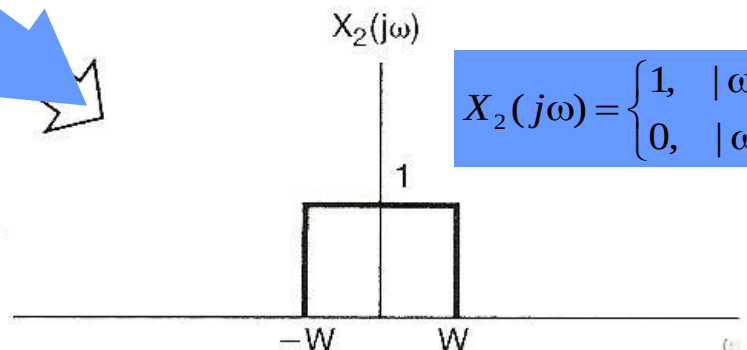


$$X_1(j\omega) = \frac{2 \sin \omega T_1}{\omega}$$

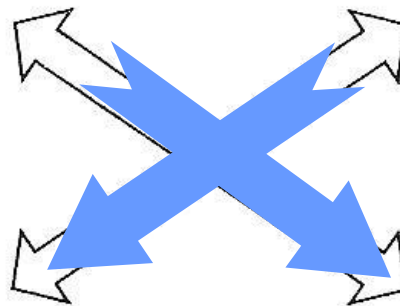
$$x_2(t) = \frac{\sin Wt}{\pi t}$$



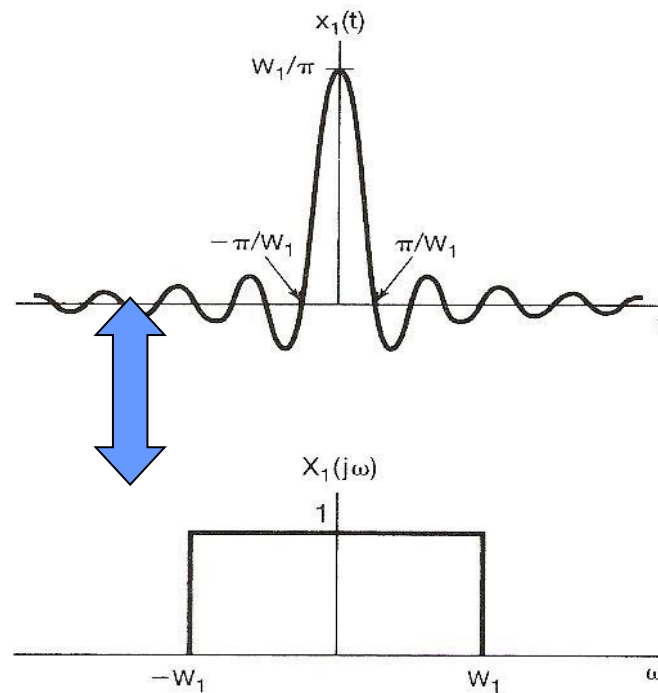
$\mathcal{F}$



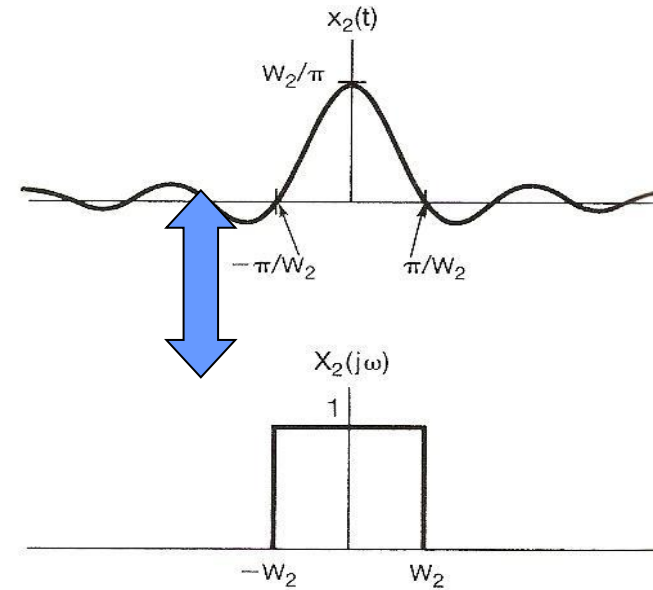
$$X_2(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$



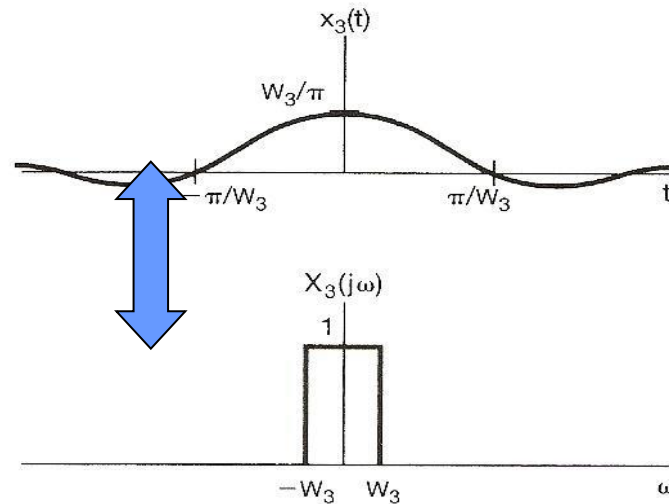
inverse
relationship
between signal
“width” in
time/frequency
domains



(a)



(b)



(c)

$$x(t) \longleftrightarrow X(j\omega)$$

CTFT Properties (cont.)

- Time reversal

$$x(-t) \longleftrightarrow X(-j\omega)$$

- Conjugate Symmetry

$$x(t) \text{ real} \longleftrightarrow X(-j\omega) = X^*(j\omega)$$

$$|X(-j\omega)| = |X(j\omega)|$$

Even

Or

$$\text{Re}\{X(-j\omega)\} = \text{Re}\{X(j\omega)\}$$

Even

$$\angle X(-j\omega) = -\angle X(j\omega)$$

Odd

$$\text{Im}\{X(-j\omega)\} = -\text{Im}\{X(j\omega)\}$$

Odd



a) $x(t)$ real and even

$$x(t) = x(-t) = x^*(t)$$

$$\Rightarrow X(j\omega) = X(-j\omega) = X^*(j\omega) \text{ — Real \& even}$$

b) $x(t)$ real and odd

$$x(t) = -x(-t) = x^*(t)$$

$$\Rightarrow X(j\omega) = -X(-j\omega) = -X^*(j\omega) \text{ — Purely imaginary \& odd}$$

$$\text{c) } X(j\omega) = \text{Re}\{X(j\omega)\} + j\text{Im}\{X(j\omega)\}$$

For real $x(t) = \underset{\uparrow}{\text{Ev}\{x(t)\}} + \underset{\uparrow}{\text{Od}\{x(t)\}}$

Table 4.2

Basic Fourier Transform Pairs

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0$, otherwise
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1$, $a_k = 0$, $k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave		
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$ for all k
$x(t) \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi \delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

$$e^{-at} u(t) \longleftrightarrow \frac{1}{a + j\omega}$$



Example 4.3 Impulse function

$$(a) \quad x(t) = \delta(t)$$

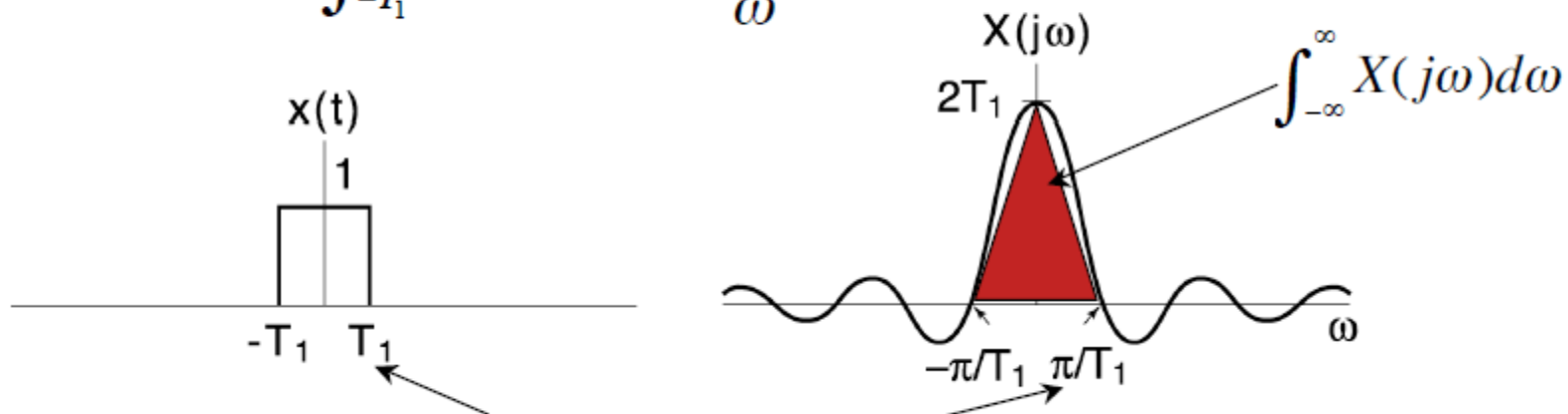
$$X(j\omega) = \int_{-\infty}^{+\infty} \delta(t) e^{-j\omega t} dt = 1$$

$$\Downarrow$$

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{j\omega t} d\omega \quad \text{— Synthesis equation for } \delta(t)$$

Example 4.4 A square pulse in the time-domain

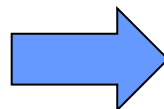
$$X(j\omega) = \int_{-T_1}^{T_1} e^{-j\omega t} dt = \frac{2 \sin \omega T_1}{\omega}$$



Note the inverse relation between the two widths $\Rightarrow$ Uncertainty principle

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$



Useful facts about CTFT's

$$X(0) = \int_{-\infty}^{+\infty} x(t) dt$$

$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) d\omega$$

- Let $x(t)$ be a signal with Fourier transform $X(j\omega)$.

Suppose we are given the following facts:

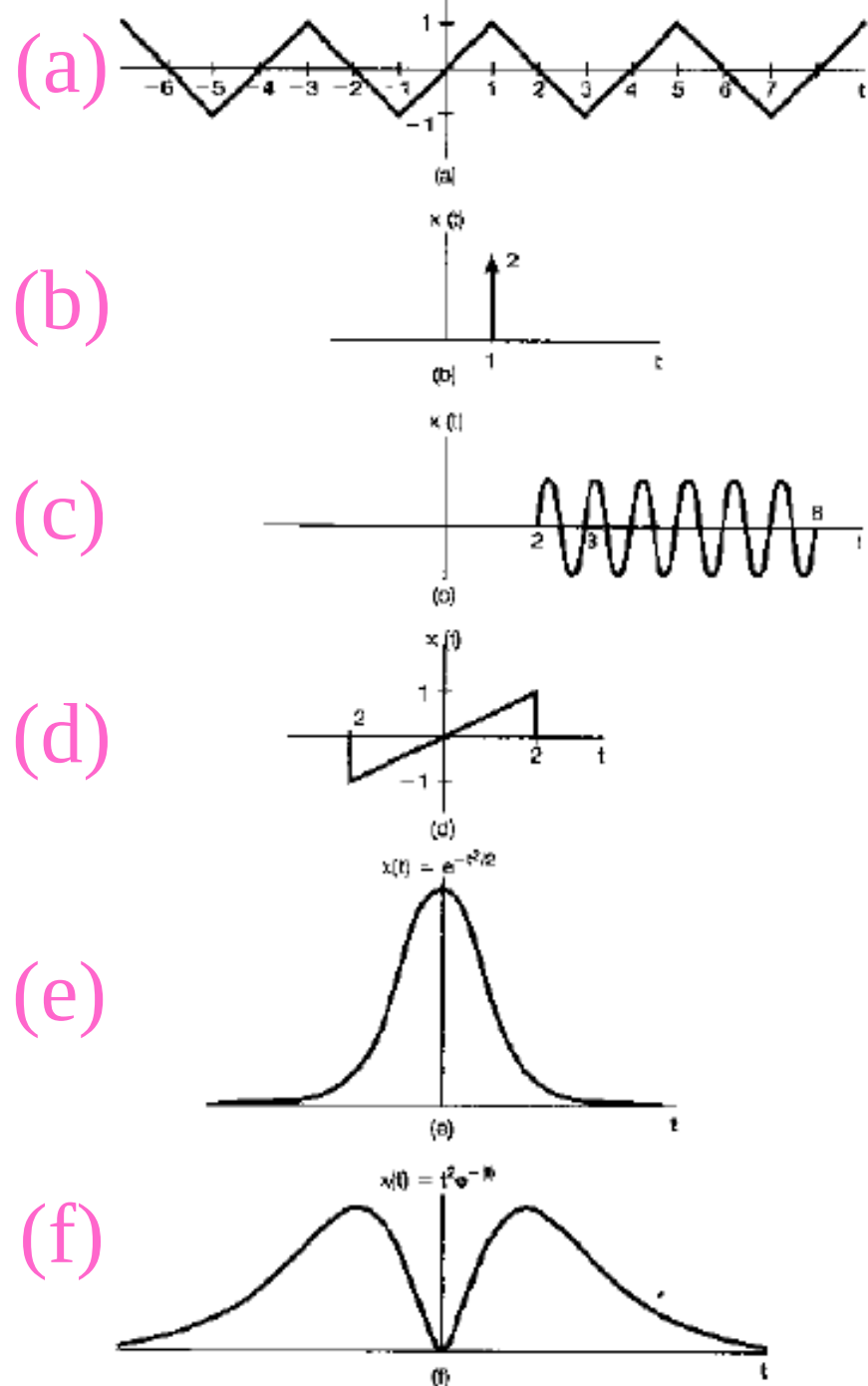
- ◆ $x(t)$ is real
- ◆ $x(t)=0$ for $t \leq 0$
- ◆ $\frac{1}{2\pi} \int_{-\infty}^{\infty} \operatorname{Re}\{X(j\omega)\} e^{j\omega t} d\omega = |t| e^{-|t|}$

Determine a closed-form expression for $x(t)$.

Problem 4.24 (a)

- Determine which, if any, of the real signals in (a)-(f) have Fourier transforms that satisfy each of the following condition:

- ◆ $\text{Re}\{X(j\omega)\} = 0$
- ◆ $\text{Im}\{X(j\omega)\} = 0$
- ◆ There exists a real a such that $e^{ja\omega} X(j\omega)$ is real
- ◆ $\int_{-\infty}^{\infty} X(j\omega) d\omega = 0$
- ◆ $\int_{-\infty}^{\infty} \omega X(j\omega) d\omega = 0$



CTFT Properties

8) Convolution Property

$$y(t) = h(t) * x(t) \longleftrightarrow Y(j\omega) = H(j\omega)X(j\omega)$$

where $h(t) \longleftrightarrow H(j\omega) \quad x(t) \longleftrightarrow X(j\omega)$

Basically a consequence of the eigenfunction property

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \quad \Rightarrow \quad x(t) = \int_{-\infty}^{+\infty} \underbrace{\left(\frac{1}{2\pi} X(j\omega) d\omega \right)}_{\text{coefficient}} e^{j\omega t}$$

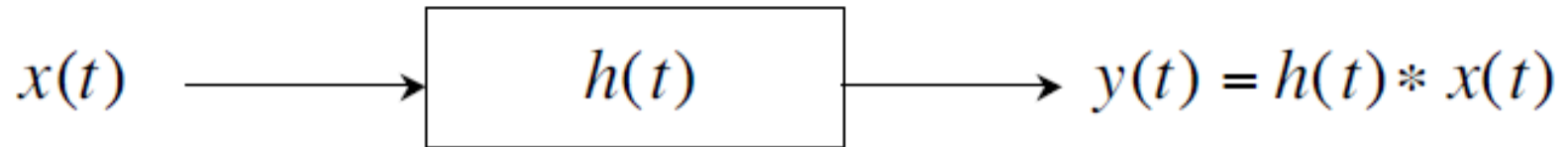
$$a e^{j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow H(j\omega) a e^{j\omega t}$$

$\Downarrow$ superposition

$$y(t) = \int_{-\infty}^{+\infty} \underbrace{\left(H(j\omega) \cdot \frac{1}{2\pi} X(j\omega) d\omega \right)}_{\text{New coefficient}} e^{j\omega t}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \underbrace{H(j\omega) X(j\omega)}_{Y(j\omega)} e^{j\omega t} d\omega$$

The Frequency Response Revisited

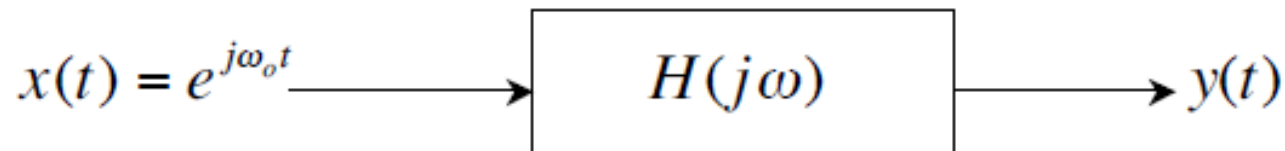


$$Y(j\omega) = H(j\omega)X(j\omega)$$



The frequency response $H(j\omega)$ of a CT LTI system is simply the Fourier transform of its impulse response $h(t)$

Example #1:



Recall

$$e^{j\omega_o t} \longleftrightarrow 2\pi\delta(\omega - \omega_o)$$

$$Y(j\omega) = H(j\omega)X(j\omega) = H(j\omega)2\pi\delta(\omega - \omega_o) = 2\pi H(j\omega_o)\delta(\omega - \omega_o)$$



$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} Y(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi H(j\omega_o) \delta(\omega - \omega_o) e^{j\omega t} d\omega$$

$$y(t) = H(j\omega_o) e^{j\omega_o t}$$

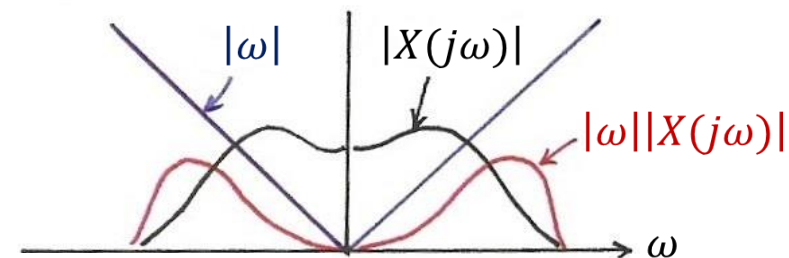
Frequency Response Examples

Example 4.16 A differentiator

$$y(t) = \frac{dx(t)}{dt} \quad \text{— an LTI system}$$

From differentiation property $\Rightarrow \frac{d}{dt} \xleftrightarrow{FT} j\omega$

$$\Downarrow \\ H(j\omega) = j\omega$$



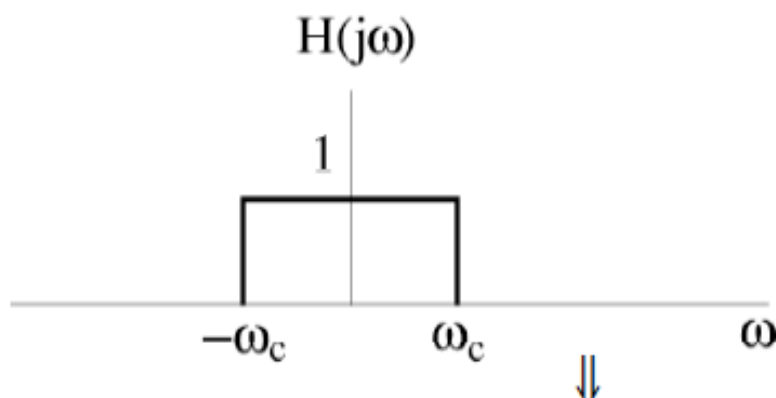
1) Amplifies high frequencies (enhances sharp edges)

2) $+\pi/2$ phase shift ($j = e^{j\pi/2}$) Larger at high ω_0 phase shift

$$\frac{d}{dt} \sin \omega_0 t = \omega_0 \cos \omega_0 t = \omega_0 \sin(\omega_0 t + \pi/2)$$

$$\frac{d}{dt} \cos \omega_0 t = -\omega_0 \sin \omega_0 t = \omega_0 \cos(\omega_0 t + \pi/2)$$

Example 4.18 : Impulse Response of an *Ideal* Lowpass Filter

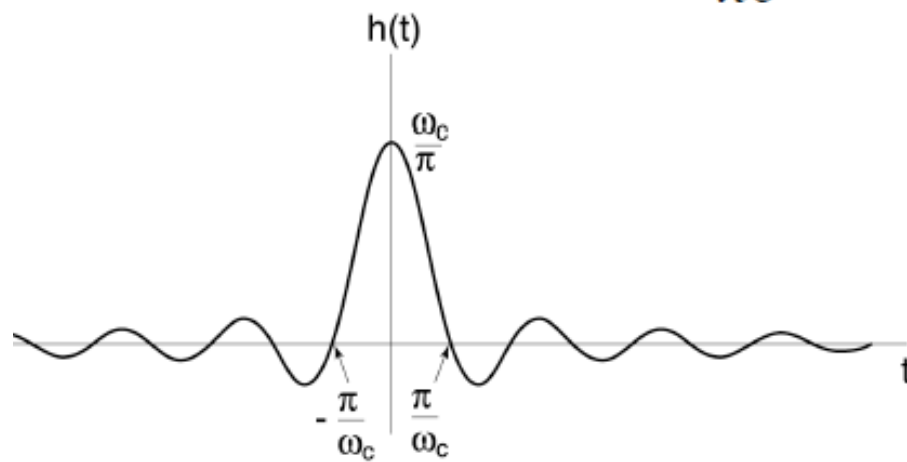


$$h(t) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega t} d\omega = \frac{\sin \omega_c t}{\pi t} = \frac{\omega_c}{\pi} \text{sinc}\left(\frac{\omega_c t}{\pi}\right)$$

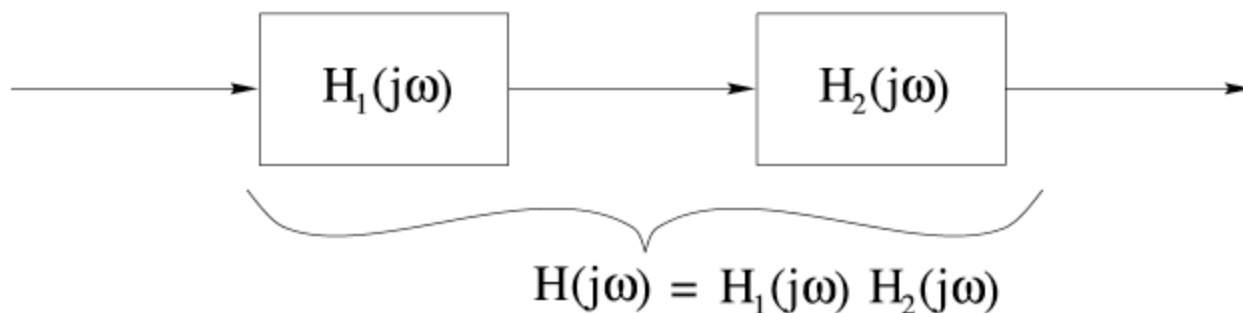
Questions:

- 1) Is this a causal system?
- 2) What is $h(0)$?

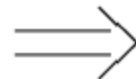
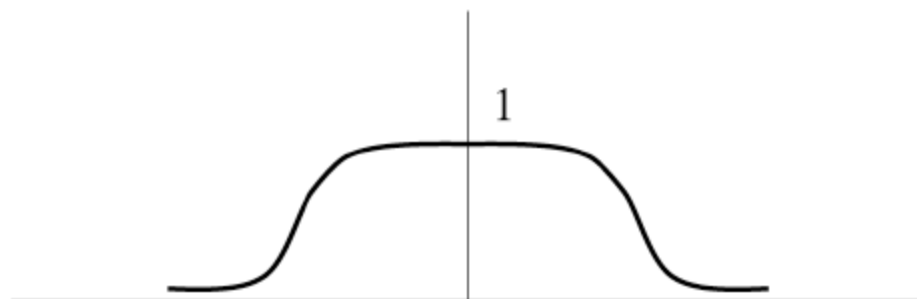
$$\text{sinc}(\theta) = \frac{\sin(\pi\theta)}{\pi\theta}$$



Example #4: Cascading filtering operations



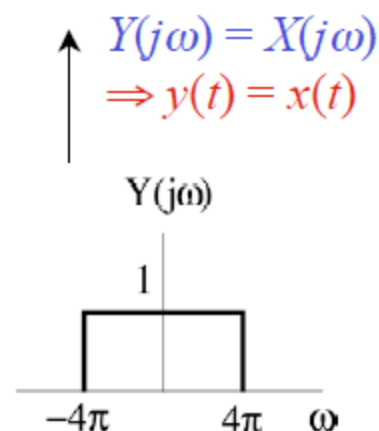
e.g. $H_1(j\omega) = H_2(j\omega)$



$H(j\omega) = H_1^2(j\omega)$ has a
sharper frequency
selectivity

Example 4.20

$$\underbrace{\frac{\sin 4\pi t}{\pi t}}_{x(t)} * \underbrace{\frac{\sin 8\pi t}{\pi t}}_{h(t)} = ?$$



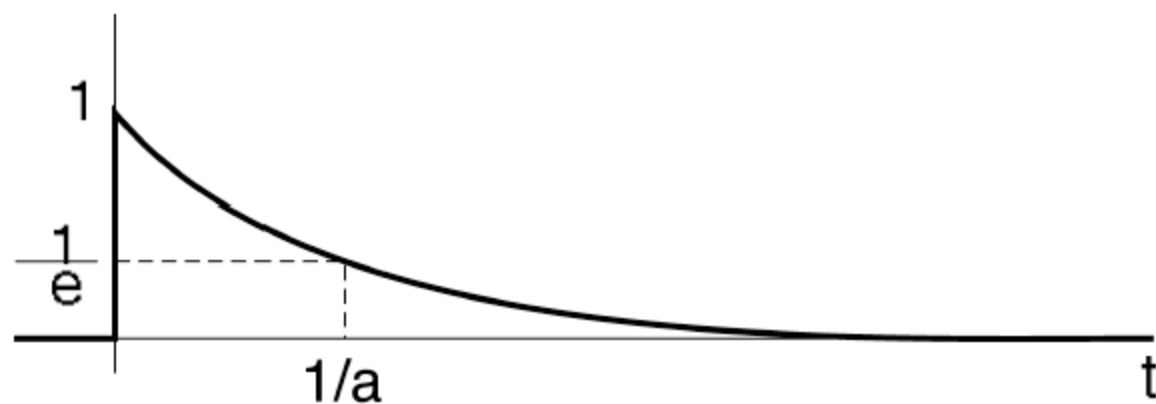
Example #6:

$$e^{-at^2} * e^{-bt^2} = ?$$

$$\begin{array}{c} \Downarrow \\ \sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}} \times \sqrt{\frac{\pi}{b}} e^{-\frac{\omega^2}{4b}} = \frac{\pi}{\sqrt{ab}} e^{-\frac{\omega^2}{4} \left(\frac{1}{a} + \frac{1}{b} \right)} \end{array}$$

Gaussian $\times$ Gaussian = Gaussian, Gaussian $*$ Gaussian = Gaussian

Review from the last lecture, right-sided exponential



$$x(t) = e^{-at}u(t) , \quad a > 0$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt = \int_0^{+\infty} \underbrace{e^{-at}e^{-j\omega t}}_{e^{-(a+j\omega)t}} dt$$

$$= -\left(\frac{1}{a+j\omega}\right)e^{-(a+j\omega)t}\bigg|_0^{\infty} = \boxed{\frac{1}{a+j\omega}}$$

Example 4.19

$$h(t) = e^{-t}u(t) \quad , \quad x(t) = e^{-2t}u(t)$$

$$y(t) = h(t) * x(t) = ?$$

$\Downarrow$

$$Y(j\omega) = H(j\omega)X(j\omega) = \frac{1}{(1+j\omega)} \cdot \frac{1}{(2+j\omega)}$$

$\Downarrow$ Partial fraction expansion

$$Y(j\omega) = \frac{1}{1+j\omega} \overset{a=1}{-} \frac{1}{2+j\omega} \overset{a=2}{+}$$

$\Downarrow$ inverse FT

$$y(t) = [e^{-t} - e^{-2t}]u(t)$$

CTFT Properties

9) Multiplication Property

Since *FT* is highly symmetric,

$$x(t) = \mathcal{F}^{-1}\{X(j\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega, \quad X(j\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

thus if

$$x(t) * y(t) \longleftrightarrow X(j\omega) \cdot Y(j\omega)$$

then the other way
around is also true

$$\begin{aligned} x(t) \cdot y(t) &\longleftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega) \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta) Y(j(\omega - \theta)) d\theta \end{aligned}$$

$\frac{1}{2\pi}$

— A consequence of *Duality*

Examples of the Multiplication Property: Modulation Property

Frequency shift

$$e^{j\omega_0 t} x(t) \xleftrightarrow{F} X(j(\omega - \omega_0))$$

$$e^{j\omega_0 t} \cdot x(t) \longleftrightarrow \frac{1}{2\pi} [2\pi\delta(\omega - \omega_0) * X(j\omega)] \\ = X(j(\omega - \omega_0))$$

Example 4.21

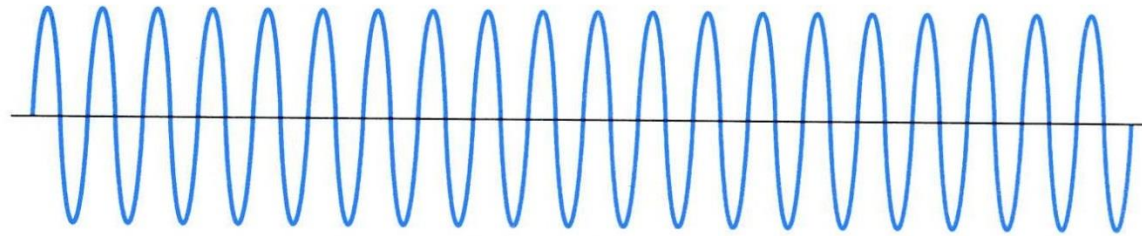
$$r(t) = s(t) \cdot p(t) \longleftrightarrow R(j\omega) = \frac{1}{2\pi} [S(j\omega) * P(j\omega)]$$

$$\text{For } p(t) = \cos\omega_0 t \longleftrightarrow P(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

$$R(j\omega) = \frac{1}{2} S(j(\omega - \omega_0)) + \frac{1}{2} S(j(\omega + \omega_0))$$

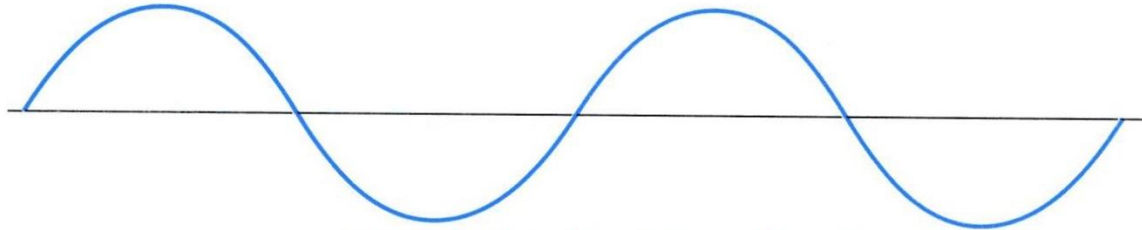
(cont.)

$p(t)$



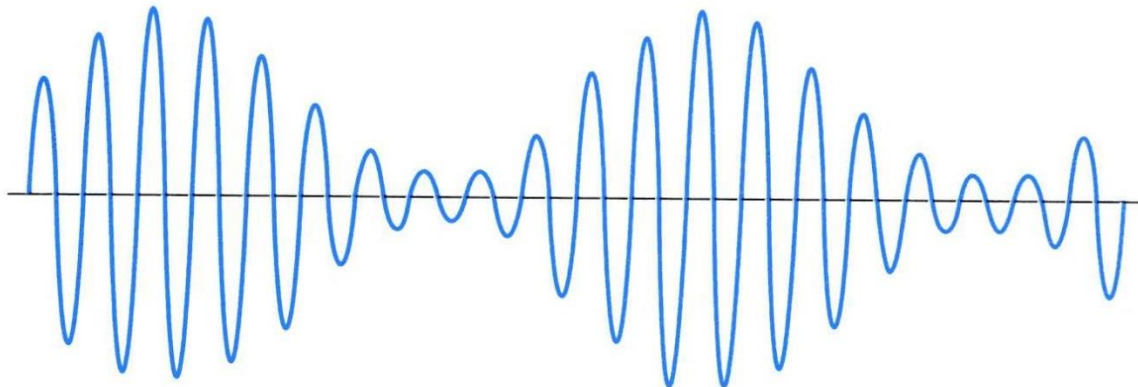
Carrier Signal

$s(t)$



Modulating Sine Wave Signal

$r(t)$

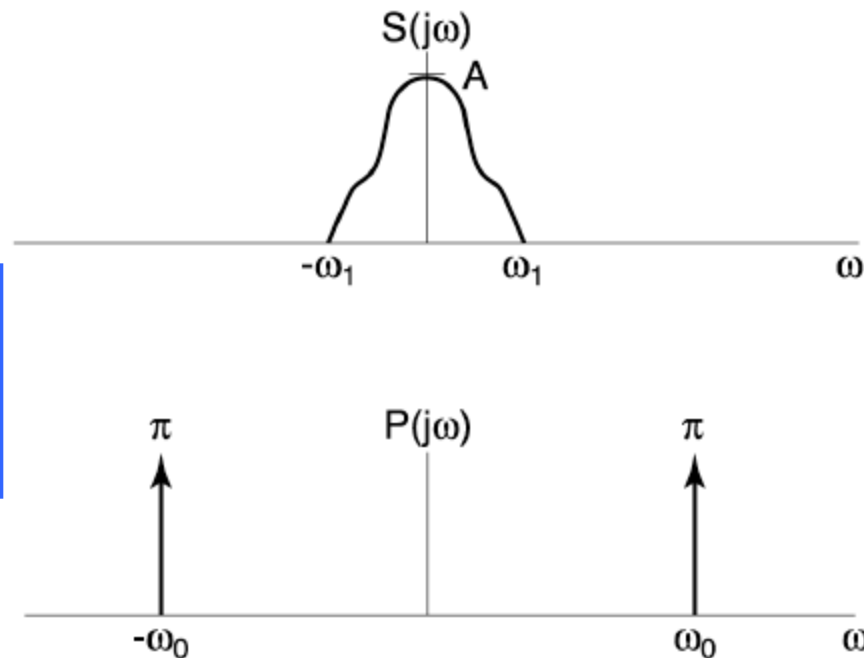


Amplitude Modulated Signal

(cont.)

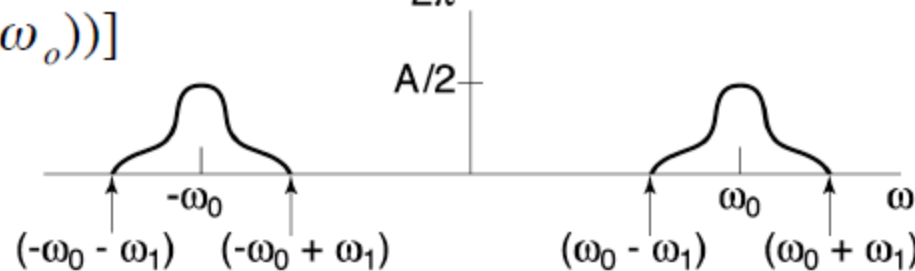
 ω_1 : bandwidth

$r(t) = s(t) \cdot \cos(\omega_o t)$
 Amplitude
 modulation (*AM*)



$$R(j\omega) = \frac{1}{2} [S(j(\omega - \omega_o)) + S(j(\omega + \omega_o))]$$

$$R(j\omega) = \frac{1}{2\pi} [S(j\omega) * P(j\omega)]$$



Drawn assume
 $\omega_o - \omega_1 > 0$
i.e. $\omega_o > \omega_1$

Frequency-Selective Filtering with Variable Center Frequency

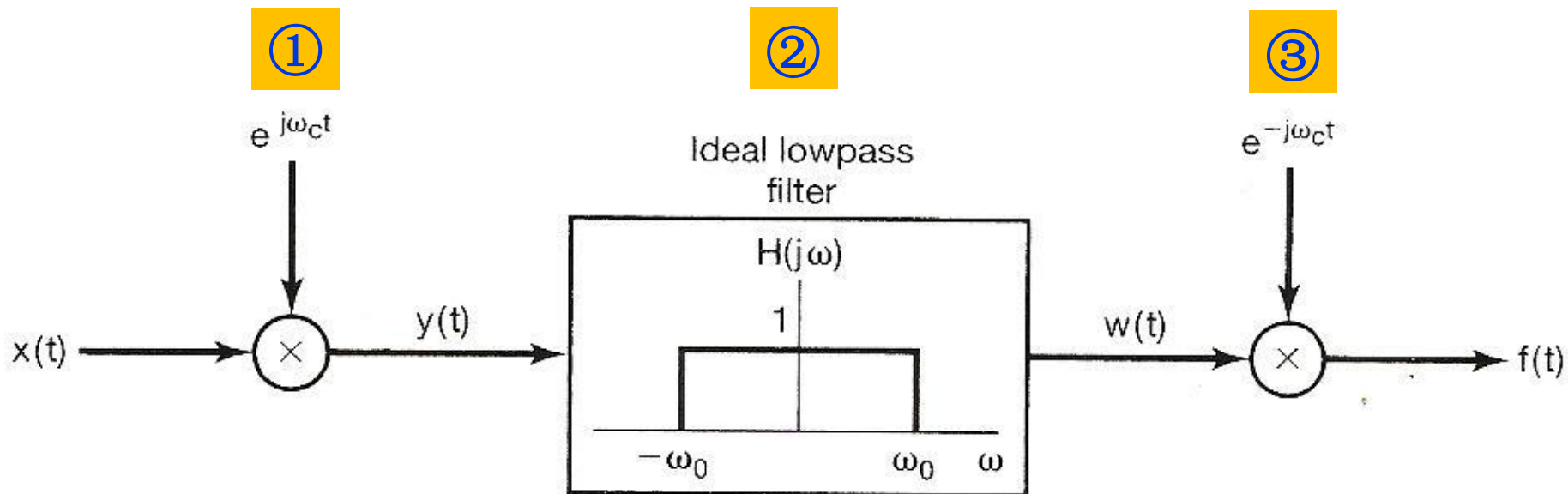
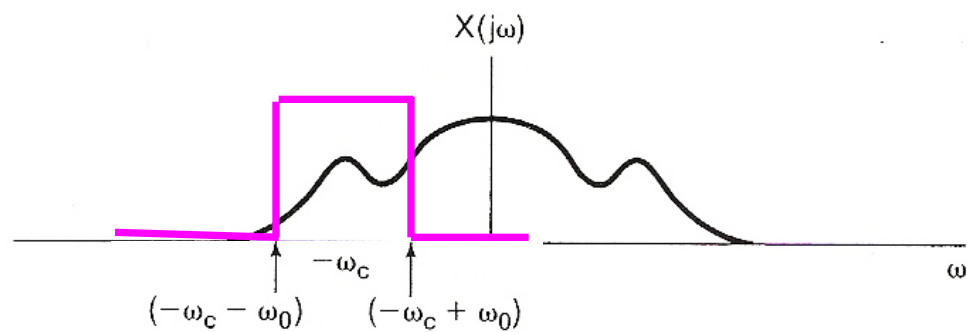
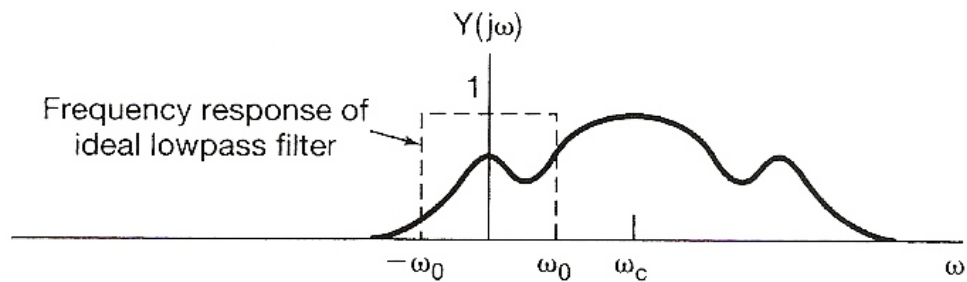


Figure 4.26 Implementation of a bandpass filter using amplitude modulation with a complex exponential carrier.

①



②



③

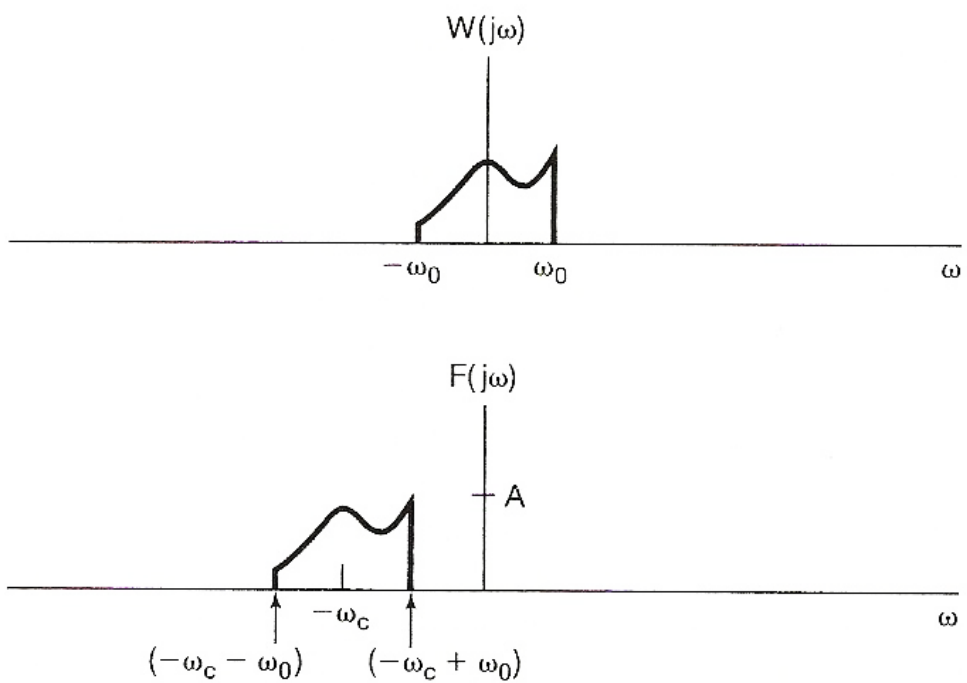


Figure 4.27 Spectra of the signals in the system of Figure 4.26.

Table 4.1

Properties of the

Fourier Transform

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$ $y(t)$	$X(j\omega)$ $Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt} x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^t x(t)dt$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re\{X(j\omega)\} = \Re\{X(-j\omega)\} \\ \Im\{X(j\omega)\} = -\Im\{X(-j\omega)\} \\ X(j\omega) = X(-j\omega) \\ \angle X(j\omega) = -\angle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}\{x(t)\}$ [x(t) real] $x_o(t) = \mathcal{O}\{x(t)\}$ [x(t) real]	$\Re\{X(j\omega)\}$ $j\Im\{X(j\omega)\}$
4.3.7	Parseval's Relation for Aperiodic Signals		
		$\int_{-\infty}^{+\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) ^2 d\omega$	

Example #8: LTI Systems Described by LCCDE's (Linear-constant-coefficient differential equations)

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

Using the Differentiation Property

$$\frac{d^k x(t)}{dt^k} \longleftrightarrow (j\omega)^k X(j\omega)$$

$\Downarrow$ Transform both sides of the equation

$$\sum_{k=0}^N a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^M b_k (j\omega)^k X(j\omega)$$

$\Downarrow$

$$Y(j\omega) = \underbrace{\left[\frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \right]}_{H(j\omega)} X(j\omega)$$

$$H(j\omega) = \left[\frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \right]$$

- A causal and stable LTI system S has the frequency response

$$H(j\omega) = \frac{j\omega + 4}{6 - \omega^2 + 5j\omega}$$

- ◆ Determine a differential equation relating the input $x(t)$ and output $y(t)$ of S
 - ◆ Determine the impulse response $h(t)$ of S
 - ◆ What is the output of S when the input is $x(t) = e^{-4t}u(t) - te^{-4t}u(t)$
-